Stark resonances under exponential-cosine screening: through-threshold maps for the quantum-plasma hydrogen atom

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(Dated: July 24, 2026)

The exponential-cosine screened Coulomb (ECSC) potential of dense quantum plasmas differs from Debye screening in one structural respect: its tail oscillates. We compute the resonance spectrum of the hydrogen atom under ECSC screening combined with a static electric field, using a coupled-channel complex-scaling method whose screened blocks cost nothing beyond the Debye machinery — the ECSC potential is the half-sum of two Yukawa potentials at complex screening parameters, which also yields the analytic rotation bound $\theta_c < \pi/4$. Three results organize the physics. At zero field, parametric continuation answers a question left open in the literature: between the critical screening $\lambda_c = 0.7205$ and the birth of tail shape resonances at $\lambda \approx 0.90$, the ground-state pole merges into the rotated continuum — consistent with the conjectured virtual-state scenario — while the $2p$ pole detaches at $\lambda_c(2p) = 0.1482$ as a genuine resonance, rotation-stable to 10^{-13}. With the field on, we map the ground resonance on a 5×19 grid of the (F, λ) plane: at every field strength the resonance crosses $E_r = 0$ near $\lambda^* \approx 0.68$, almost independently of the field, and continues smoothly as a positive-energy resonance — a regime with no Debye analog in the same parameter range, reached here because the field keeps the pole isolated where the field-free pole dissolves. Finally, comparing ECSC and Debye screening at matched binding energy isolates the tail effect on ionization: the oscillatory tail suppresses the field-ionization width by roughly a factor of two at weak field, the suppression weakening as the field strengthens and the barrier peak moves inward.

I. INTRODUCTION

Screening in a dense, degenerate electron gas is not exponential. Quantum hydrodynamic treatments of test-charge screening in quantum plasmas [1] lead to the exponential-cosine screened Coulomb (ECSC) potential,

$$V(r) = -\frac{Z}{r}\, e^{-\lambda r} \cos(\lambda r), \tag{1}$$

whose study as a model interaction long predates that derivation [2]. The oscillating factor is not a cosmetic difference from the Debye–Hückel form: the potential changes sign, its tail alternates between attraction and repulsion, and this modifies both the bound spectrum — critical screening parameters differ substantially [3] — and the resonance structure. Field-free ECSC resonances have recently been mapped with high precision by the complex-scaling pseudospectral work of Jiao and collaborators [4], who located the loss of the ground state at $\lambda_c = 0.720524$ and the emergence of well-defined s-wave shape resonances at $\lambda_{r+} = 0.90035$, and who explicitly note that the behavior in the intermediate window is unexplained, conjecturing conversion into virtual states. Perturbative field responses (polarizabilities and hyperpolarizabilities) of ECSC-screened atoms are likewise known [5], and the one nonperturbative Stark calculation under ECSC screening — complex-scaled resonance energies and widths at selected parameter points — was carried out by Wang, Chen, and Guo [6]. What does not exist is systematic coverage: a resonance map of the combined field–screening plane, the fate of excited Stark manifolds, or a controlled comparison against Debye screening that isolates what the oscillating tail actually does to field ionization. This paper supplies the three.

The tool is the coupled-channel complex-scaling method of our Debye companion study [7], and the extension to ECSC screening is free: Eq. (1) is the half-sum of two Yukawa potentials at complex screening parameters $\lambda(1 \pm i)$, so every screened matrix element reuses the Debye code path at complex argument, and the identity yields at once the analytic bound $\theta_c < \pi/4$ on the rotation angle. All runs are verified against exact limits at run time, and the Yukawa mode of the present engine reproduces the companion engine to 2×10^{-16}.

II. METHOD

The $m = 0$ coupled-channel complex-scaling machinery is that of Ref. [7]: each angular channel carries the orthonormal generalized-Laguerre radial basis with common scale $\beta = 0.35$, $n_b = 70$ functions per channel and $\ell_{\max} = 14$ (1050 degrees of freedom) at production settings, a shared 3000-point Gauss–Legendre quadrature, dipole blocks $(\ell + 1)/\sqrt{(2\ell + 1)(2\ell + 3)}$ coupling adjacent channels, and analytic scaling of every operator [8–12]; the Stark term is covered by the Herbst–Simon theory [13, 14]. The single ECSC-specific element is the screened factor: writing $e^{-\lambda r} \cos \lambda r = \frac{1}{2}[e^{-\mu_+ r} + e^{-\mu_- r}]$ with $\mu_\pm = \lambda(1 \pm i)$, the complex-scaled screening enters as two Yukawa evaluations at $\mu_\pm r e^{i\theta_c}$. Decay of these

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factors requires $\mathrm{Re}(\mu_\pm e^{i\theta_c}) > 0$, i.e.

$$\theta_c < \pi/4, \tag{2}$$

a hard analytic cap (enforced in the code) comfortably above the production angles $\theta_c = 0.18/0.28$ (retry 0.30/0.42). Resonances are identified exactly as in Ref. [7]: parametric continuation in λ with the search window predicted from the $F = 0$ bound curve plus the previous complex field shift, acceptance by two-angle agreement with the modulus difference recorded as the spread, and θ_c-trajectories computed for the least favorable points. Uncertainties quoted are these spreads — error estimates calibrated against the anchors below, not bounds.

Verification. The hydrogen limit reproduces $-1/2n^2$ to 7×10^{-11} (a failed check aborts every run); the $\lambda = 0$ row of the production map reproduces the pure-Stark anchors of Ref. [7] (themselves validated to ten digits against the classical Stark values [15–17]); the small-λ expansion of the ground level follows the exact law $E_{1s} = -\frac{1}{2} + \lambda - \lambda^3\langle r^2\rangle/3 + O(\lambda^4)$ — the λ^2 term is absent for ECSC, unlike Debye screening — to the expected order (deviation 1.2×10^{-8} at $\lambda = 0.01$); and our critical screening $\lambda_c(1s) = 0.72027$ agrees with the high-precision value 0.720524 of Ref. [3] within the threshold-diffusivity bias of a variational basis, documented in Ref. [18]. No published ECSC Stark table was available to us in full [6]; a point-by-point comparison remains for the submission stage.

III. RESULTS

A. Field-free pole structure: the intermediate window

Reference [4] leaves open what happens between $\lambda_c = 0.720524$, where the ground state ceases to be bound, and $\lambda_{r+} = 0.90035$, where sharp s-wave shape resonances appear. Parametric continuation answers this directly. Tracking the $1s$ pole across λ_c with fine steps, the pole moves smoothly and basis-stably into the lower half plane, but its rotation response degrades monotonically: the two-angle spread grows from 5×10^{-5} at $\lambda = 0.715$ to 10^{-2} at $\lambda = 1.25$, no θ_c-plateau develops, and — the decisive diagnostic — the tracked ratio $(\Gamma/2)/E_r$ converges to $\tan 2\theta_c$ of the rotation angle used, the signature of the rotated-continuum ray rather than of an isolated pole. Within uniform complex scaling, then, the $1s$ pole does not survive as an isolated resonance in the intermediate window: it merges into the continuum, exactly as the virtual-state scenario conjectured in Ref. [4] implies. (The sharp tail resonances above λ_{r+} live too close to the rotated ray for the present uniform-scaling setup; we do not attempt to reproduce them and refer to Ref. [4].) We record the ratio test itself as a practical artifact discriminator: any near-threshold candidate whose width-to-position ratio tracks $\tan 2\theta_c$, or whose parameters are insensitive to λ, is a discretized ray state, not a resonance.

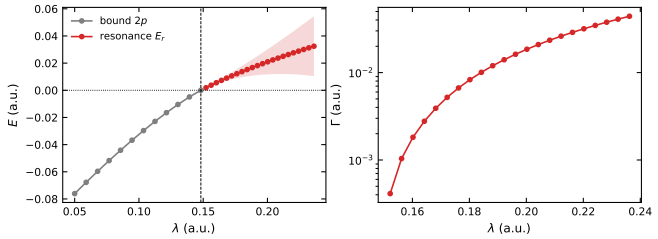


FIG. 1. Field-free $2p$ pole under ECSC screening: bound branch (gray), and the shape resonance born at $\lambda_c(2p) = 0.148205$ (red; band = $\pm\Gamma/2$). Right: width on a log scale.

The $\ell = 1$ story is the clean counterpart (Fig. 1). The $2p$ level unbinds at $\lambda_c(2p) = 0.148205$ — earlier than the Debye value 0.220217 [3, 18], the oscillating tail screens harder at distance — and detaches as a genuine shape resonance behind the centrifugal-plus-tail barrier: near threshold the two-angle spread is at machine level (10^{-13}), and the resonance broadens rapidly as λ grows, reaching $\Gamma \sim 5 \times 10^{-2}$ by $\lambda = 0.24$.

B. The ground-state (F, λ) map: through the threshold

Figure 2 shows the ground-state resonance on the grid $F \in \{0.02, \ldots, 0.10\} \times \lambda \in \{0, 0.05, \ldots, 0.90\}$: all 95 points tracked by continuation with zero losses, two-angle spreads below 1.5×10^{-4} at the broadest corner and below 10^{-7} over 58% of the grid (machine-readable tables provided). Table I lists a subset.

The headline is the left panel: at every field strength the ground resonance crosses $E_r = 0$ at $\lambda^* \approx 0.68$ — remarkably field-independent (from 0.686 at $F = 0.02$ to 0.681–0.682 at stronger fields, a shallow minimum near $F = 0.06$; total variation below 0.01) — and continues smoothly to positive energies, reaching $E_r = +0.049$, $\Gamma = 0.097$ at $F = 0.02$, $\lambda = 0.9$. For $F \neq 0$ the crossing is not a spectral event — the Stark operator's spectrum covers the real line and the scaled operator's eigenvalues continue analytically [13] — but its contrast with the field-free limit is physical and sharp: at $F = 0$ the same pole merges into the continuum upon crossing (Sec. III A), while any finite field keeps it isolated, because the field-ionization channel gives the pole a finite width scale of its own and detaches it from the threshold. The through-threshold ground resonance also has no Debye analog on the same grid: under Yukawa screening the ground E_r remains negative over the entire map of Ref. [7] (its field-free level, still bound at $\lambda = 0.8$ with $E = -0.045$, only approaches threshold at the much larger critical screening 1.19). Positive-energy ground resonances under extreme Debye screening were tabulated by Yu and Ho [19]; the ECSC potential brings

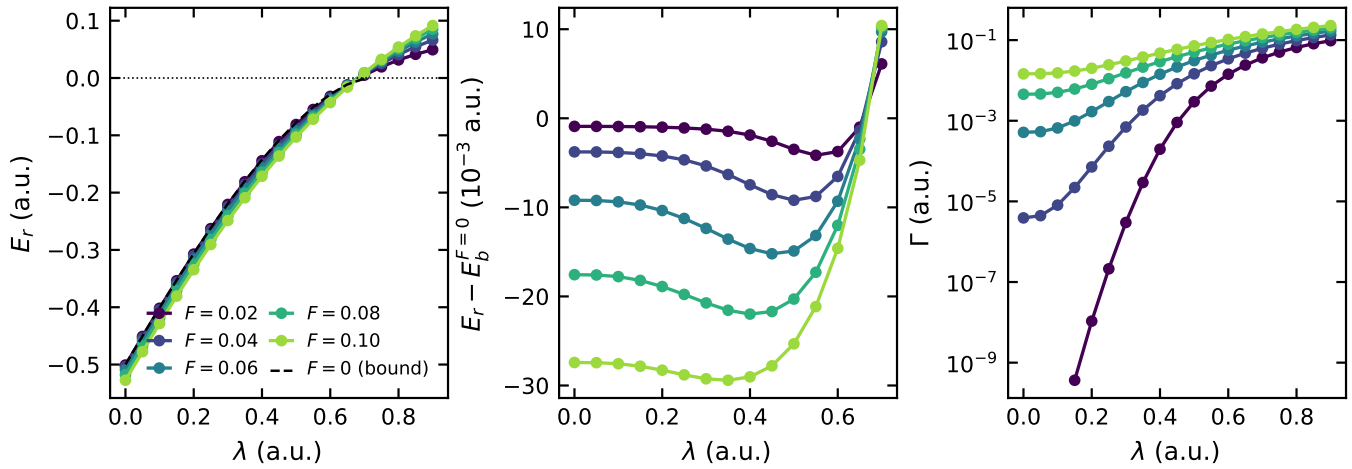


FIG. 2. Ground-state resonance across the (F, λ) plane under ECSC screening. Left: position E_r (dashed: $F = 0$ bound level; dotted: $E_r = 0$). Center: field-induced displacement relative to the bound level (plotted where the $F = 0$ level exists). Right: width (log scale; points shown where Γ exceeds ten times the spread).

TABLE I. Ground-state resonance parameters (a.u.) at selected screenings. Widths marked $< 10^{-10}$ are below the resolved level. λ^* is the $E_r = 0$ crossing (linear interpolation, ± 0.01).

F	λ	E_r	Γ	λ^*
0.02	0.3	-0.220638	3.00×10^{-6}	0.686
	0.6	-0.032042	1.43×10^{-2}	
	0.9	$+0.049297$	9.66×10^{-2}	
0.04	0.3	-0.224766	7.00×10^{-4}	0.681
	0.6	-0.034837	3.43×10^{-2}	
	0.9	$+0.066174$	1.35×10^{-1}	
0.06	0.3	-0.231777	5.27×10^{-3}	0.681
	0.6	-0.037618	5.62×10^{-2}	
	0.9	$+0.076982$	1.68×10^{-1}	
0.08	0.3	-0.240144	1.54×10^{-2}	0.681
	0.6	-0.040327	7.95×10^{-2}	
	0.9	$+0.085034$	1.99×10^{-1}	
0.10	0.3	-0.248664	3.04×10^{-2}	0.682
	0.6	-0.042906	1.04×10^{-1}	
	0.9	$+0.091514$	2.30×10^{-1}	

that regime down to screening parameters well inside the physically relevant range.

Two secondary features mirror the Debye map [7] and one does not. Screening again catalyzes field ionization (right panel: at $F = 0.02$ the width climbs eight orders of magnitude across the grid), and the perturbative Stark regime again shrinks with screening. The non-mirroring feature is the field displacement (center panel): it turns *positive* as $\lambda \to \lambda_c$, i.e. near the critical screening the field *raises* the resonance position relative to the vanishing bound level — a threshold effect absent from the Debye map at comparable screenings.

TABLE II. Ground Stark widths at matched field-free binding E_b. $\lambda_{\rm Y}, \lambda_{\rm E}$: screening parameters at which Yukawa/ECSC give the same E_b (10^{-9} match).

E_b	$\lambda_{\rm Y}$	$\lambda_{\rm E}$	F	$\Gamma_{\rm E}/\Gamma_{\rm Y}$
-0.40	0.10823	0.10091	0.04	0.551
			0.08	0.823
-0.30	0.23699	0.20697	0.04	0.447
			0.08	0.677
-0.20	0.39703	0.32394	0.04	0.493
			0.08	0.648
-0.12	0.56482	0.43240	0.04	0.571
			0.08	0.660

C. Debye versus ECSC at matched binding: what the oscillating tail does to ionization

The comparison that isolates the tail effect must remove the trivial difference — at equal λ the two potentials bind differently. We therefore match the *binding*: for each anchor E_b we solve for $\lambda_{\rm Y}$ (Yukawa) and $\lambda_{\rm E}$ (ECSC) such that the field-free ground level equals E_b to 10^{-9}, then compare the Stark widths at equal F. Table II shows the result: at matched binding the ECSC width is systematically *smaller* — by about a factor of two at $F = 0.04$ ($\Gamma_{\rm E}/\Gamma_{\rm Y} = 0.45$–$0.55$) and less at $F = 0.08$ (0.65–0.82). The quasistatic exponent is identical by construction, so the suppression is a pure barrier-shape effect: beyond its first zero at $r = \pi/2\lambda_{\rm E}$ the ECSC potential stops assisting the field in lowering the Stark barrier (and briefly opposes it), so the effective barrier is wider than its Coulomb-tailed Yukawa counterpart. As F grows the barrier peak moves inward, into the region where the two potentials coincide, and the suppression

fades — both trends visible in the table.

D. The $n = 2$ doublet

The $m = 0$ doublet at $F = 0.02$ (parabolic labels $k = \pm 1$ assigned by continuation from $\lambda = 0$, not quantum numbers) behaves as under Debye screening but on a compressed λ scale, consistent with $\lambda_c(2p)$: the uphill component crosses $E_r = 0$ at $\lambda \approx 0.094$ (Debye: 0.112 [7]) and continues to $E_r = +0.127$, $\Gamma = 0.079$ at $\lambda = 0.30$; the downhill width roughly triples over the same range ($0.061 \to 0.208$), with the ratio Γ_{-1}/Γ_{+1} drifting $1.97 \to 2.63$. Spreads are below 10^{-5} except at the four broadest points (worst 6×10^{-3} for the downhill component at $\lambda = 0.30$, which we quote to two digits only).

IV. CONCLUSIONS

The ECSC extension of the coupled-channel complex-scaling program cost one identity and delivered three results: the resolution of the intermediate-window question of Ref. [4] (pole merging, with a practical $\tan 2\theta_c$ artifact discriminator as a byproduct), the first systematic (F, λ) resonance map under ECSC screening — whose ground state crosses into a positive-energy resonance at accessible screenings, with the field acting as the isolating agent — and the matched-binding comparison showing the oscillating tail suppresses field ionization by up to a factor of two, a pure barrier-shape effect that fades with increasing field. The machinery extends unchanged to excited manifolds, $m \neq 0$, and other screening families; the relativistic (Dirac) version of the combined problem is the natural next step and is under way.

ACKNOWLEDGMENTS

The author thanks the ESPOCH Faculty of Sciences for institutional support. All scripts, gates, and machine-readable grids are distributed with the companion programs and will be archived with a persistent identifier at publication.

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